ViBridge-VQA: Bridging Global and Local Visual Tokens for Parameter-Efficient Vietnamese Visual Question Answering

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Abstract. Vision–language models (VLMs) such as Vintern-1B bring strong multimodal abilities to Vietnamese, but adapting them to a new Visual Question Answering (VQA) benchmark usually means fine-tuning on many high-resolution image tiles or training a new architecture. We propose **ViBridge-VQA**, which improves Vintern-1B without updating any of its pre-trained weights. From a single image view, the frozen decoder receives *global* tokens, produced by a small trainable bridge from the `[CLS]` vector, and *local* tokens, produced by Vintern-1B's own pre-aligned projector and pooled to a budget chosen as the knee of the accuracy–cost Pareto front. On the reasoning-oriented AutoViVQA benchmark, ViBridge-VQA trains 1.35% of the parameters and reaches an F1 of 54.6, comparable to the fine-tuned Vintern-1B, while its decoder reads 36 times fewer visual tokens; a lightweight decoder LoRA raises F1 to 58.2. Reusing the pre-trained projector proves more effective than learning a new connector, and the local tokens complement decoder adaptation and transfer to three other Vietnamese VQA datasets.

Keywords: Vietnamese Visual Question Answering · Vision–Language Models · Parameter-Efficient Learning · Visual Token Budget · Generative VQA

1 Introduction

Visual Question Answering (VQA) asks a model to answer a natural-language question about an image. The task has progressed rapidly with large vision–language models (VLMs) such as BLIP-2 [12], Flamingo [1] and LLaVA [19], which connect a pre-trained vision encoder to a pre-trained language model and generate free-form answers. Vietnamese has followed this trend: Vintern-1B [7] connects an InternViT-300M encoder [6] to a Qwen2.5-0.5B decoder [26] and is an open Vietnamese VLM, while benchmarks such as ViVQA [29] and the recent

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reasoning-oriented AutoViVQA [21] make it possible to measure progress in this low-resource setting.

Obtaining a strong model on a new Vietnamese benchmark, however, is still expensive. The common recipe fine-tunes an existing VLM: the fine-tuned Vintern-1B baseline of AutoViVQA adds LoRA adapters to the language model and splits every image into up to six high-resolution tiles plus a thumbnail, so that the decoder reads up to 1,792 visual tokens per image. The alternative is to design a task-specific architecture, as ViMoE-VQA [20] does with a sparse mixture-of-experts fusion module and a decoder trained on AutoViVQA. A cheaper option is to keep the whole VLM frozen and train only a small connector between its encoder and decoder. Existing connectors follow two designs, each with its own weakness. A *global-only bridge* maps the global image vector to a few tokens; it is cheap, but it loses local details such as small objects, printed text and spatial relations. A *learned patch compressor*, such as a Q-Former, keeps these details, but it has to learn a new mapping from visual features to the decoder, which is difficult with limited Vietnamese data.

Vintern-1B suggests a third option. Its projector was pre-trained to map image regions into the input space of the decoder, so this alignment already exists and does not need to be learned again. At the same time, the model discards the global `[CLS]` vector of its encoder, which summarises the whole scene. This raises our research question: *when labelled data is limited, is reusing the pre-trained visual alignment of a VLM, with pooling to control the number of tokens, more effective than learning a new connector, and how many global and local tokens does the frozen decoder need?*

To address this question, we propose **ViBridge-VQA**, which gives the frozen decoder of Vintern-1B two complementary kinds of visual tokens from a single image view. *Local* tokens reuse the frozen projector of Vintern-1B, average-pooled to a chosen budget and placed in its original image slot, so the decoder receives region features in the representation it was trained to read. *Global* tokens are produced from the `[CLS]` vector by a small trainable bridge and add the scene-level information that Vintern-1B ignores. Only this bridge is trained. Because every additional token increases inference cost, we treat the number of global and local tokens as a budget and select it as the knee of the accuracy–cost Pareto front. As a result, ViBridge-VQA matches the published fine-tuned Vintern-1B while its decoder reads far fewer visual tokens and none of its weights is updated.

In summary, our main contributions are as follows:

- We propose **ViBridge-VQA**, a global–local bridge that improves a frozen Vietnamese VLM by combining trainable global tokens with local tokens from its own pre-trained projector, and that can be combined with a lightweight decoder LoRA.
- We treat the number of visual tokens as an explicit **accuracy–cost budget** and select it on the validation split as the knee of the Pareto front.
- We **analyse where the gains come from**: local tokens matter far more than global ones, reusing the pre-trained projector works better than learning

a new connector, local tokens complement decoder adaptation, and they transfer to other Vietnamese VQA datasets.

2 Related Work

Early VQA systems classified over a fixed set of answers, whereas current generative VLMs produce free-form answers with an autoregressive language model [12,1,19]. For Vietnamese, available VQA datasets include ViVQA [29], the synthetic ViCLEVR [31], the book-cover ViOCRVQA [25] and the scene-text ViTextVQA [23]. AutoViVQA [21] contributes 37,077 questions on 19,411 COCO images [17], generated by large language models, filtered by an ensemble of validators and annotated with reasoning categories, with an emphasis on relational, causal and multi-step reasoning. On the modelling side, BARTPhoBEiT [30] and AViVQA-TranConI [22] combine pre-trained visual and Vietnamese language representations, Vintern-1B [7] brings an instruction-tuned VLM to Vietnamese, and ViMoE-VQA [20] routes multimodal tokens to sparse experts for generative VQA on AutoViVQA. AutoViVQA and ViMoE-VQA are earlier work of our group; here we use the AutoViVQA data but no component of ViMoE-VQA, and the method, the image-disjoint split and all our results are new. These approaches either train a new model or fine-tune an existing VLM on multi-tile inputs; in contrast, we keep every pre-trained weight of Vintern-1B frozen and use a single image view.

Keeping the vision encoder and the language model frozen and training only a connector is a common way to build VLMs efficiently, and connectors differ mainly in what they must learn. Frozen [32] learns a visual prefix from a global image representation. The Q-Former of BLIP-2 [12] learns a set of queries that compress the patch features, which preserves details but requires a large module (188M parameters) pre-trained on millions of image–text pairs. LLaVA [19,18] projects every patch token with a small MLP, which preserves locality but passes hundreds of tokens to the decoder; Honeybee [4] proposes locality-preserving abstractors, and InternVL [6], on which Vintern-1B is built, merges neighbouring patches before its MLP projector. In all of these models, the mapping from patches to the decoder is learned during large-scale pre-training. ViBridge-VQA differs in two respects: it does not learn this local mapping again but reuses the projector already pre-trained in the VLM, and it trains only a small global bridge (12.9M parameters) on about 26k task questions.

The number of visual tokens largely determines the cost of the decoder. Inference-optimal scaling of VLMs [13] shows that a few tokens can suffice for many tasks, and Matryoshka multimodal models [3] train a model to work at several nested pooled token budgets and show that the required budget depends on the task. Other work reduces the visual tokens directly: DeCo [34] replaces the Q-Former with adaptive average pooling of the patch tokens and reports better results, TokenPacker [14] learns a compact projector, and LLaVA-PruMerge [28] and FastV [5] prune or merge visual tokens of an existing model. Like DeCo, we rely on simple average pooling, but we apply it to the output of a projector that

is already pre-trained, keep the whole VLM frozen, and choose among budgets with explicit criteria, namely the knee of the Pareto front [27] and TOPSIS [11]. Parameter-efficient methods such as adapters [9], prefix tuning [15] and LoRA [10] reduce the cost of adaptation from the other side, by training a small number of additional weights in the decoder. We use decoder LoRA both as a baseline and as a complement, and ask whether a small training budget is better spent on the visual tokens or on adapting the decoder.

3 Proposed Method

We formulate VQA as conditional answer generation: given an image I and a Vietnamese question q, the model generates a short free-form answer $y = (y_1, \ldots, y_T)$. ViBridge-VQA builds on Vintern-1B [7] and keeps all of its pre-trained weights frozen. Instead of adapting the model, it changes what the decoder sees.

3.1 Overall Architecture

As illustrated in Fig. 1, ViBridge-VQA keeps the three components of Vintern-1B: the InternViT-300M vision encoder (24 layers, width 1024), its two-layer MLP projector P, and the Qwen2.5-0.5B-Instruct language decoder (24 layers, width 896). Each image is resized to one 336×336 view (24×24 patches of 14×14 pixels); InternViT, pre-trained at 448×448, bicubically interpolates its position embeddings to this resolution, which roughly halves the encoder cost (191 instead of 362 GMACs per view). The encoder returns one vector per patch and a global `[CLS]` vector $v \in \mathbb{R}^{1024}$. Vintern-1B uses only the patch vectors: it merges every 2×2 block of neighbouring patches into one 4096-dimensional vector (pixel shuffle) and maps each merged vector into the decoder input space with P, which was pre-trained to align the two backbones. At our resolution this yields a 12×12 grid of 144 region tokens. ViBridge-VQA uses both outputs of the encoder.

3.2 Global Tokens

The `[CLS]` vector summarises the whole scene, but Vintern-1B never passes it to the decoder. A small trainable bridge g_ϕ expands it into g tokens with two linear heads:

$$g_\phi(v) = \left[\, W_a v + b_a;\ \mathrm{reshape}_{(g-1)\times 896}(W_e v + b_e) \,\right] \in \mathbb{R}^{g \times 896}. \tag{1}$$

where $W_a \in \mathbb{R}^{896\times 1024}$ and $W_e \in \mathbb{R}^{(g-1)896\times 1024}$. The first head produces an *anchor* token, and the second produces $g-1$ complementary tokens that present the global content to the decoder from different learned directions. The bridge has $1025 \cdot 896 \cdot g$ parameters, i.e., 12,857,600 for $g = 14$, and its g tokens are placed before the prompt. We consider $g \in \{8, 14\}$: $g = 8$ is our Global-only model, and $g = 14$ gave the highest F1 when the Global-only bridge was varied from 4 to 20 tokens (Section 4.5). The final pair (g, k) is selected with the criterion of Section 3.5.

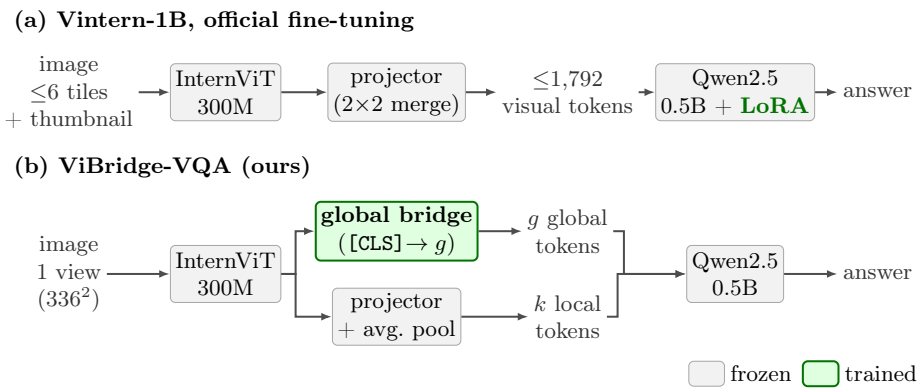


Fig. 1: Overview of ViBridge-VQA compared with the official fine-tuning of Vintern-1B. (a) The official recipe encodes up to seven views and adapts the decoder with LoRA. (b) ViBridge-VQA encodes one view; a trainable bridge maps the `[CLS]` vector to g global tokens, and the frozen projector gives k pooled local tokens (selected: $g = 14$, $k = 36$).

3.3 Local Tokens

To give the decoder region-level detail, ViBridge-VQA also passes local tokens, taken directly from the 144 region tokens of the frozen projector P. To control their number, we average-pool the 12×12 grid to an $s \times s$ grid, so that $k = s^2$ with $s \in \{1, 3, 6, 12\}$: $k = 144$ keeps every region token, and $k = 36$ averages each 2×2 block. The pooled tokens replace the embeddings of k image-context placeholders inside Vintern-1B's own image slot, `<img>` $\cdots$ `</img>`, in the prompt. Because they come from the pre-trained projector rather than from a newly learned projection of the patches, the local tokens stay in the representation that the decoder was trained to read, and no parameters are needed to produce them.

3.4 Training Objective

Only the bridge parameters ϕ are trained; the vision encoder, the projector and the decoder remain frozen, and no adapter is added to the decoder. We minimise the cross-entropy of the answer tokens:

$$\mathcal{L} = -\sum_{t=1}^{T} \log p\big(y_t \mid y_{<t},\, g_\phi(v),\, \mathrm{pool}_s(P(x)),\, q\big), \tag{2}$$

where x denotes the merged patch vectors. ViBridge-VQA can also be combined with decoder adaptation: in that variant, a rank-16 LoRA [10] on the query, key, value and output projections of every decoder layer is trained jointly with the bridge, which adds 2,162,688 parameters. Throughout the paper, percentages of trained parameters are relative to all parameters of the model, i.e., the frozen components of Vintern-1B (about 0.94B, including the vision encoder, projector, embeddings and output layer) plus the trained modules. With $g = 14$, the bridge

alone trains 12.86M parameters (1.35%), and the bridge with LoRA 15.02M (1.57%).

3.5 Choosing the Token Budget

The cost of the decoder grows with the number of visual tokens $g + k$, so we treat the pair (g, k) as a budget to be chosen. In our experiments, F1 increases with every added token among the configurations with local tokens, so all of them lie on the accuracy–token Pareto front and the front alone does not single one out. We therefore select its *knee* [27], the point after which additional tokens bring diminishing returns. After normalising cost c and accuracy a to $[0, 1]$ over the front, the knee is the configuration that maximises $\hat{a} - \hat{c}$, i.e., the one lying furthest above the line joining the cheapest and the most accurate configuration. The knee is computed on the validation split only, and we check the choice with the multi-criteria method TOPSIS [11], also on validation. Test results are computed for every configuration only to be reported; no choice is based on them.

4 Experiments

After comparing ViBridge-VQA with existing systems (Section 4.3), we answer four questions: whether reusing the pre-trained projector is better than learning a new connector (Section 4.4), how many visual tokens are needed (Section 4.5), whether local tokens complement decoder adaptation (Section 4.6), and whether the gains transfer to other datasets (Section 4.7).

4.1 Dataset and Evaluation Metrics

Dataset. We evaluate on AutoViVQA [21], a Vietnamese VQA benchmark on MS COCO images [17] in which each question has five short free-form reference answers and a reasoning type. To measure answers about images the model has never seen, we use an image-disjoint split. After removing 273 duplicated (image, question) pairs and 97 questions without one of the eight reasoning labels, we partition the 36,707 questions by image into a 70/15/15 split (seed 42), so that no image appears in more than one split: 25,776 / 5,463 / 5,468 questions on 13,579 / 2,909 / 2,910 images. In a preliminary comparison, this split lowered the F1 of Global-only by about 0.5 points relative to a split that does not group questions by image.

Evaluation metrics. Following AutoViVQA and ViMoE-VQA, we report exact-match Accuracy, word-level Precision, Recall and F1, BLEU [24], ROUGE-L [16], METEOR [2] and CIDEr [33], all against the five reference answers and multiplied by 100. All our models are scored with the same implementation.

4.2 Experimental Setup

Implementation details. The vision encoder, the projector and the decoder are initialised from Vintern-1B-v3.5 [7] and kept frozen. ViBridge-VQA is trained for two epochs, after which neither the validation loss nor CIDEr improves further; the variants with a decoder LoRA are trained for one epoch, and Global-only + LoRA also for three epochs, its strongest setting (+1.3 validation F1). We use AdamW with a learning rate of 2×10^{-4} and an effective batch size of 8, train on the first reference answer, and decode greedily (batch size 2, at most 50 new tokens). Unless stated otherwise, every configuration is trained with three random seeds (42, 123 and 3407), and we report the mean and the standard deviation. Every model fits on a single 16 GB GPU such as an NVIDIA T4. The token budget is selected on the validation split, and the test split is used only for the final report.

Baselines. We compare against three groups of systems: (1) multimodal models fine-tuned on AutoViVQA, namely ViT5_ViT, BARTPhoBEiT [30], Vintern-1B and ViMoE-VQA [20]; (2) models used zero-shot, namely Vintern-1B, LLaMA 3.2, Gemini 2.0 and 2.5 Flash, and GPT-5; and (3) our *Global-only* model, the Multi-Token bridge of Eq. (1) with $g = 8$ and no local tokens, with and without a decoder LoRA (*Global-only + LoRA*). Our models are *ViBridge-VQA* ($g = 14$, $k = 36$), *ViBridge-VQA + LoRA* and *ViBridge-VQA (k=144)*. The results of groups (1) and (2) are those reported in the AutoViVQA and ViMoE-VQA papers. The fine-tuned Vintern-1B follows the official recipe: the vision encoder and the projector are frozen, a rank-16 LoRA adapts the language model for one epoch, and each image is encoded as up to six 448×448 tiles plus a thumbnail.

4.3 Main Results

Table 1 compares ViBridge-VQA with fine-tuned multimodal models and zero-shot language models. Without updating any weight of Vintern-1B, ViBridge-VQA raises its F1 from 17.55 to 54.56, a level comparable to the published fine-tuned Vintern-1B (53.76), while its decoder reads 50 instead of 1,792 visual tokens. Only its exact-match accuracy is lower (11.75 versus 13.01).

Decoder adaptation further improves these results. Training a rank-16 LoRA jointly with the bridge for one epoch raises F1 to 58.17 and gives the highest exact-match accuracy in Table 1. ViMoE-VQA, a task-specific model whose fusion module and decoder are trained on AutoViVQA, keeps the highest precision and F1, whereas ViBridge-VQA reuses an existing VLM and trains only 1.57% of its parameters. With the same one-epoch LoRA, ViBridge-VQA is 4.7 validation F1 above Global-only (58.21 versus 53.47; Table 2). Our models are stable across seeds: on the test split, the standard deviation is at most 0.19 for F1 and 0.95 for CIDEr, and test F1 is at most 0.6 points below validation F1.

Table 1: Performance on the AutoViVQA test set (×100). Baselines: results reported in AutoViVQA [21] and ViMoE-VQA [20] (ViMoE-VQA: mean of five seeds). Ours: ViBridge-VQA ($g = 14$, $k = 36$, 50 visual tokens) without and with a one-epoch decoder LoRA, image-disjoint split (Section 4.1), mean over three seeds with the standard deviation below, 1.35% and 1.57% of the parameters trained. Bold: best per column.

	Model	Accuracy	Precision	Recall	F1	BLEU	ROUGE-L	METEOR	CIDEr
Fine-tuned	ViT5_ViT	7.97	46.84	50.33	48.52	4.13	46.89	31.02	72.68
	BARTPhoBEiT	8.81	45.30	46.48	45.88	4.33	44.83	24.57	**188.96**
	Vintern-1B	13.01	52.47	55.12	53.76	6.11	51.93	35.25	72.84
	ViMoE-VQA	9.55	**62.89**	58.60	**60.69**	12.54	47.12	39.10	88.76
Zero-shot	Vintern-1B	0.12	17.52	19.87	17.55	1.91	25.84	23.93	8.54
	LLaMA 3.2	0.36	23.96	73.71	36.16	3.62	36.11	30.01	62.84
	Gemini 2.0 F.	0.55	27.20	74.10	39.79	4.41	39.60	31.72	74.42
	Gemini 2.5 F.	0.22	24.43	**76.66**	24.75	0.39	37.27	31.22	71.90
	GPT-5	10.84	47.20	55.20	50.89	6.07	47.30	33.34	84.20
Ours	ViBridge-VQA	11.75 ±0.21	55.36 ±0.21	56.56 ±0.21	54.56 ±0.19	18.52 ±0.18	52.64 ±0.19	45.53 ±0.33	110.58 ±0.95
	ViBridge-VQA + LoRA	**14.67** ±0.30	59.02 ±0.12	59.96 ±0.33	58.17 ±0.19	**22.53** ±0.11	**56.25** ±0.22	**49.32** ±0.37	122.88 ±0.93

Table 2: Source of the visual tokens, without and with a decoder LoRA (validation, ×100, mean ± std over three seeds; Multi-Token: four seeds; *one seed). Params: trained bridge parameters. Best results per block are in bold.

Visual tokens from	Tokens	Params	F1	BLEU	ROUGE-L	METEOR	CIDEr
Bridge only (two epochs)							
Residual	1	4.86M	46.02 ±.36	12.98 ±.15	44.23 ±.38	36.86 ±.31	86.48 ±.77
Multi-Token	8	7.35M	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
Tile-Attention	8	4.14M	45.14 ±.92	12.34 ±.32	43.28 ±.95	36.11 ±.73	84.14 ±1.40
Light Q-Former	8	27.6M	47.06 ±.66	13.99 ±.33	45.37 ±.63	37.57 ±.60	88.07 ±1.89
Full Q-Former	16	69.4M	46.84 ±.47	13.76 ±.36	45.10 ±.41	37.43 ±.34	87.74 ±1.24
ViBridge-VQA (k=36)	50	12.86M	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01
+ decoder LoRA (one joint epoch, 2.16M parameters)							
Residual	1	4.86M	52.69 ±.14	18.94 ±.36	51.04 ±.13	43.33 ±.33	104.25 ±.93
Multi-Token	8	7.35M	53.47 ±.17	19.72 ±.09	51.79 ±.15	44.11 ±.23	106.68 ±.69
Tile-Attention*	8	4.14M	52.82	19.13	51.25	43.44	104.72
Light Q-Former	8	27.6M	53.21 ±.14	19.92 ±.31	51.54 ±.13	43.77 ±.19	106.26 ±.73
Full Q-Former	16	69.4M	53.29 ±.29	19.57 ±.25	51.62 ±.30	43.96 ±.21	106.50 ±1.04
ViBridge-VQA (k=36)	50	12.86M	**58.21** ±.16	**22.27** ±.17	**56.42** ±.14	**49.55** ±.06	**124.88** ±.75

4.4 Is Reusing the Projector Better than Learning a New Connector?

Yes. Table 2 compares three sources of visual tokens: the global `[CLS]` vector (a one-token *Residual* bridge or the *Multi-Token* bridge of Eq. (1)), a learned compression of the patches (*Tile-Attention* with eight learned queries, or a light or full Q-Former [12]), and our local tokens from the pre-trained projector. Without decoder adaptation, the bridges that learn to compress the patches stay 3.7–5.6 F1 below the Multi-Token bridge despite having up to 9.4× more parameters, whereas adding 36 or 144 local tokens from the projector to the same bridge ($g = 8$) raises F1 by 3.6 and 5.4 points (Table 3). Training Global-only to imitate

Table 3: Effect of the token budget on the validation set (×100, mean ± std over three seeds; Global-only: four seeds). Tokens: visual tokens passed to the decoder; Time: minutes to evaluate the validation and test questions on one Tesla T4. The selected budget is in bold.

g	k	Tokens	Time	F1	BLEU	ROUGE-L	METEOR	CIDEr
8	0	8	–	50.74 ±.17	16.88 ±.23	49.03 ±.12	41.38 ±.21	99.07 ±.47
14	0	14	–	50.67 ±.08	16.81 ±.22	48.91 ±.11	41.34 ±.23	99.83 ±.53
8	1	9	57.0	51.03 ±.12	16.54 ±.27	49.21 ±.13	41.37 ±.11	99.27 ±.58
8	9	17	59.2	52.82 ±.11	17.58 ±.03	50.96 ±.05	43.56 ±.20	106.11 ±.55
8	36	44	60.6	54.32 ±.41	18.22 ±.39	52.48 ±.43	45.31 ±.42	111.88 ±1.43
8	144	152	71.2	56.15 ±.08	19.82 ±.00	54.21 ±.06	47.51 ±.05	119.05 ±.25
14	1	15	58.8	51.67 ±.25	17.30 ±.36	49.93 ±.24	42.36 ±.19	101.66 ±1.11
14	9	23	57.7	52.98 ±.08	17.83 ±.23	51.17 ±.10	43.71 ±.13	106.02 ±.80
14	**36**	**50**	**60.0**	**54.96** ±.28	**18.73** ±.23	**53.10** ±.27	**46.01** ±.42	**113.29** ±1.01
14	144	158	70.2	56.61 ±.04	20.18 ±.13	54.70 ±.05	48.00 ±.05	120.71 ±.12

the projector instead, either by aligning the mean of its output tokens with that of the projector's tokens (cosine, feature distillation) or by matching the decoder's answer distribution obtained with the projector's tokens (KL, output distillation), gives 50.35 and 50.46 F1, slightly below its 50.74.

4.5 How Many Visual Tokens Are Needed?

About 50 visual tokens, 14 global and 36 local, give the best compromise between accuracy and cost. Table 3 and Fig. 2a vary the number of global tokens $g \in \{8, 14\}$ and of local tokens $k \in \{1, 9, 36, 144\}$. Local tokens have by far the larger effect: at $g = 14$, going from 1 to 144 local tokens adds 4.9 F1 and 19.1 CIDEr on validation, whereas going from 8 to 14 global tokens adds only 0.2–0.6 F1 at a fixed k. Accuracy keeps increasing with k, but the gain per token falls from 0.07 F1 between 23 and 50 tokens to 0.015 between 50 and 158 tokens, so additional tokens bring diminishing returns.

Among the configurations with local tokens, F1 grows with every added token, so all of them lie on the accuracy–token Pareto front, and we select its knee (Section 3.5). On validation, the knee is $g = 14$, $k = 36$ for every metric, and also when cost is measured as evaluation time, although times differ by only a few minutes below $k = 144$ and include model loading. TOPSIS with equal weights on accuracy and cost makes the same choice; the knee on the test split, computed only as a post-hoc check, is the same. This budget is a compromise rather than the most accurate configuration: $k = 144$ is 1.7 F1 higher, but the selected budget obtains 72% of its gain over Global-only with 32% of its tokens.

Number of global tokens. We varied the Multi-Token bridge from $g = 4$ to $g = 20$ tokens (three seeds each). Validation F1 rose from 48.5 at $g = 4$ to 49.6 at $g = 8$ and 50.7 at $g = 14$, and stayed between 50.4 and 50.6 for $g = 16$–20.

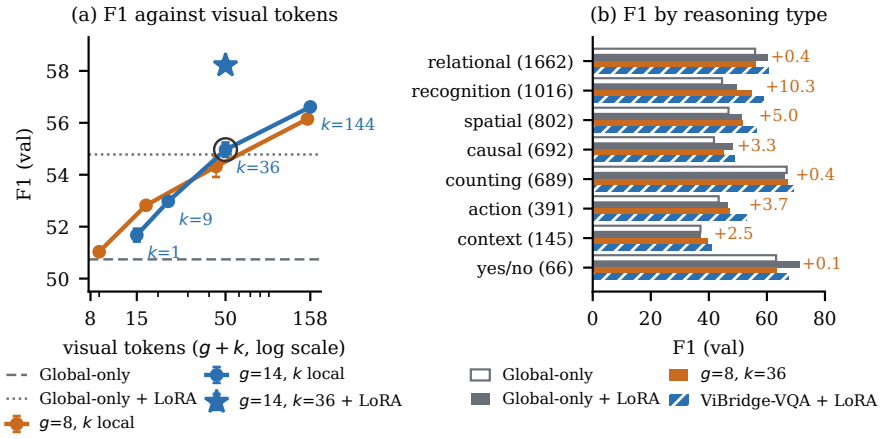


Fig. 2: (a) Validation F1 against the number of visual tokens (mean ± std over three seeds); the circled point is the knee, $g = 14$, $k = 36$, and the star ViBridge-VQA + LoRA. (b) Validation F1 by reasoning type (number of questions in parentheses), averaged over seeds; the numbers give the gain of adding 36 local tokens at the same $g = 8$. White: Global-only; grey: Global-only + LoRA; orange: $g = 8$ with local tokens; blue: $g = 14$ (ViBridge-VQA).

Computational cost. ViBridge-VQA reduces the cost in two places. Its decoder reads 50 visual tokens, whereas the official tiling (at most six tiles plus a thumbnail) splits every AutoViVQA test image into seven views, i.e., 1,792 tokens. Its vision encoder processes one 336×336 view (577 tokens, about 191 GMACs) instead of seven 448×448 views (362 GMACs each), 13 times less computation. Within our own budgets, 144 local tokens instead of 36 make evaluation about 17% slower (Table 3).

4.6 Do Local Tokens Complement Decoder Adaptation?

Yes: local tokens mainly improve perception, a decoder LoRA mainly improves reasoning answers, and the two gains add up. A decoder LoRA improves every bridge without local tokens and brings them within 0.8 F1 of one another (52.69–53.47). Even after this adaptation, all of them remain below ViBridge-VQA with a frozen decoder. Decoder adaptation is therefore not needed to surpass them, but it is still useful: in the same one-epoch joint training, it adds 3.3 F1 to ViBridge-VQA, more than the 2.7 F1 it adds to the Multi-Token bridge.

Reasoning types. Fig. 2b breaks the validation F1 down by reasoning type. Local tokens help most when the answer depends on what is visible in the image: at $g = 8$, adding 36 local tokens raises F1 by 10.3 on recognition, 5.0 on spatial and 3.7 on action questions, where this frozen-decoder model also outperforms Global-only + LoRA. Decoder adaptation helps more on causal and relational questions, whose answers are the longest (5.0 and 5.6 words on average), and on



Fig. 3: Qualitative examples from the AutoViVQA validation set (seed 42). Q: question, GT: one of the five reference answers; Ours: ViBridge-VQA ($g = 14$, $k = 36$), Ours+LoRA: ViBridge-VQA with a one-epoch decoder LoRA.

yes/no questions. ViBridge-VQA with LoRA combines both strengths and is the best of the four models in seven of the eight categories, all except yes/no.

Error analysis. Our answers are as long as the references (4.3 words on average), and most of them (77–81%) partially overlap the closest reference. Compared with Global-only, ViBridge-VQA raises the share of exact matches (F1 = 1) from 9.7% to 13.3% and lowers the share of answers with no word in common with any reference from 9.6% to 8.0% (6.8% with LoRA).

Qualitative examples. Fig. 3 illustrates how the two components work together. In every example, ViBridge-VQA already identifies the relevant content of the image (the driver's phone, the helmets, the football match, the bird feeder), but its answer is vague or incomplete. Adding the decoder LoRA turns it into the precise answer, for instance that the driver is using *the phone* and that the helmets are worn *to protect their heads*, in line with the gains of decoder adaptation on causal and yes/no questions described above.

4.7 Do the Gains Transfer to Other Datasets?

Yes on general-domain questions, while reading scene text requires more local tokens. Table 4 reports results on three Vietnamese VQA datasets whose questions were not used for training: the scene-text dataset ViTextVQA [23] and the

Table 4: Results on Vietnamese VQA datasets whose questions were not used for training (×100, mean ± std over three random samples of about 1,000 test questions; our models: seed 42; Global-only + LoRA: three epochs). Best results per dataset are in bold.

Model	F1	BLEU	ROUGE-L	METEOR	CIDEr
ViTextVQA					
Vintern-1B, zero-shot	32.01 ±.31	4.94 ±.18	20.89 ±.25	26.50 ±.54	**195.62** ±2.35
Global-only + LoRA	3.96 ±.09	0.15 ±.01	3.74 ±.12	2.61 ±.16	15.64 ±.92
ViBridge-VQA + LoRA	15.39 ±.39	1.03 ±.15	14.41 ±.33	11.17 ±.26	70.07 ±1.33
ViBridge-VQA (k=144)	**33.26** ±.67	**6.24** ±.48	**31.65** ±.73	**27.33** ±.93	190.30 ±6.71
ViVQA-X					
Vintern-1B, zero-shot	14.52 ±.47	0.06 ±.01	8.98 ±.32	11.96 ±.41	50.13 ±2.83
Global-only + LoRA	15.89 ±.65	0.18 ±.02	11.94 ±.49	9.08 ±.57	42.12 ±2.92
ViBridge-VQA + LoRA	27.64 ±.11	0.33 ±.05	21.98 ±.13	**19.24** ±.14	**95.29** ±1.51
ViBridge-VQA (k=144)	**27.89** ±.13	**0.39** ±.13	**22.02** ±.26	18.55 ±.11	92.22 ±2.31
ViVQA					
Vintern-1B, zero-shot	23.90 ±.24	0.27 ±.02	15.78 ±.25	17.84 ±.19	78.63 ±2.07
Global-only + LoRA	31.48 ±1.15	0.25 ±.09	31.48 ±1.16	21.92 ±.82	84.94 ±2.87
ViBridge-VQA + LoRA	44.76 ±1.21	**0.63** ±.02	44.74 ±1.22	36.98 ±1.28	183.60 ±7.17
ViBridge-VQA (k=144)	**44.95** ±.90	0.59 ±.17	**44.95** ±.90	**37.28** ±.61	**191.28** ±2.17

general-domain ViVQA-X [8] and ViVQA [29]. ViVQA and ViVQA-X are built on COCO images; about 10% of the sampled questions concern images that also appear, with other questions, in our training split. We evaluate our final model, ViBridge-VQA + LoRA with 50 visual tokens, and ViBridge-VQA (k=144). Both clearly outperform Global-only and, on the general-domain datasets, the zero-shot Vintern-1B (F1 44.8 and 45.0 against 23.9 on ViVQA). On ViTextVQA, the 50-token model reaches 15.4 F1, whereas ViBridge-VQA (k=144) reaches 33.3, on par with Vintern-1B, which reads up to seven 448×448 views instead of one 336×336 view.

5 Conclusion

In this work, we introduced **ViBridge-VQA**, which improves the Vietnamese VLM Vintern-1B for generative VQA without updating any of its pre-trained weights. Our results show that, for a VLM whose projector is already aligned with its decoder, reusing this projector with a small number of pooled local tokens is more effective than learning a new connector from limited data, that the number of tokens can be chosen with an explicit accuracy–cost criterion, and that the local tokens complement decoder adaptation rather than replace it.

Our study also has limitations. We train on a single dataset, AutoViVQA, and the task-specific ViMoE-VQA still reaches a higher F1. Without decoder adaptation, the exact-match accuracy of ViBridge-VQA stays below that of the fine-tuned Vintern-1B, and with 50 visual tokens it reads scene text much worse than Vintern-1B; accuracy had not saturated at the 144 local tokens available at our resolution.

Overall, an existing Vietnamese VLM can be improved substantially by choosing which visual tokens its frozen decoder receives, and how many. The code

and the image-disjoint split are available at `https://github.com/nguyennn263/modular-vlm-finetune`.

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References

1. Alayrac, J.B., Donahue, J., Luc, P., Miech, A., Barr, I., Hasson, Y., Lenc, K., Mensch, A., Millican, K., Reynolds, M., et al.: Flamingo: a visual language model for few-shot learning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 23716–23736 (2022)
2. Banerjee, S., Lavie, A.: METEOR: An automatic metric for MT evaluation with improved correlation with human judgments. In: Proceedings of the ACL Workshop on Intrinsic and Extrinsic Evaluation Measures for Machine Translation and/or Summarization. pp. 65–72 (2005)
3. Cai, M., Yang, J., Gao, J., Lee, Y.J.: Matryoshka multimodal models (2024), arXiv preprint arXiv:2405.17430
4. Cha, J., Kang, W., Mun, J., Roh, B.: Honeybee: Locality-enhanced projector for multimodal LLM. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 13817–13827 (2024)
5. Chen, L., Zhao, H., Liu, T., Bai, S., Lin, J., Zhou, C., Chang, B.: An image is worth 1/2 tokens after layer 2: Plug-and-play inference acceleration for large vision-language models (2024), arXiv preprint arXiv:2403.06764
6. Chen, Z., Wu, J., Wang, W., Su, W., Chen, G., Xing, S., Zhong, M., Zhang, Q., Zhu, X., Lu, L., Li, B., Luo, P., Lu, T., Qiao, Y., Dai, J.: InternVL: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 24185–24198 (2024)
7. Doan, K.T., Huynh, B.G., Hoang, D.T., Pham, T.D., Pham, N.H., Nguyen, Q.T.M., Vo, B.Q., Hoang, S.N.: Vintern-1B: An efficient multimodal large language model for Vietnamese (2024), arXiv preprint arXiv:2408.12480
8. Duong, T.B., Tran, H.M., Le-Nguyen, B.N., Duong, D.T.: An automated pipeline for constructing a Vietnamese VQA-NLE dataset. In: Proceedings of the International Conference on Intelligent Systems and Networks (ICISN). pp. 164–173. Lecture Notes in Networks and Systems, Springer (2026)
9. Houlsby, N., Giurgiu, A., Jastrzebski, S., Morrone, B., de Laroussilhe, Q., Gesmundo, A., Attariyan, M., Gelly, S.: Parameter-efficient transfer learning for NLP. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 2790–2799 (2019)
10. Hu, E.J., Shen, Y., Wallis, P., Allen-Zhu, Z., Li, Y., Wang, S., Wang, L., Chen, W.: LoRA: Low-rank adaptation of large language models. In: Proceedings of the International Conference on Learning Representations (ICLR) (2022)
11. Hwang, C.L., Yoon, K.: Multiple Attribute Decision Making: Methods and Applications. Springer (1981)
12. Li, J., Li, D., Savarese, S., Hoi, S.: BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In: Proceedings of the International Conference on Machine Learning (ICML). pp. 19730–19742 (2023)

13. Li, K.Y., Goyal, S., Semedo, J.D., Kolter, J.Z.: Inference optimal VLMs need only one visual token but larger models (2024), arXiv preprint arXiv:2411.03312
14. Li, W., Yuan, Y., Liu, J., Tang, D., Wang, S., Qin, J., Zhu, J., Zhang, L.: TokenPacker: Efficient visual projector for multimodal LLM (2024), arXiv preprint arXiv:2407.02392
15. Li, X.L., Liang, P.: Prefix-tuning: Optimizing continuous prompts for generation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 4582–4597 (2021)
16. Lin, C.Y.: ROUGE: A package for automatic evaluation of summaries. In: Proceedings of the ACL Workshop on Text Summarization Branches Out. pp. 74–81 (2004)
17. Lin, T.Y., Maire, M., Belongie, S., Bourdev, L., Girshick, R., Hays, J., Perona, P., Ramanan, D., Zitnick, C.L., Dollár, P.: Microsoft COCO: Common objects in context. In: Proceedings of the European Conference on Computer Vision (ECCV). pp. 740–755 (2014)
18. Liu, H., Li, C., Li, Y., Lee, Y.J.: Improved baselines with visual instruction tuning. In: Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR). pp. 26296–26306 (2024)
19. Liu, H., Li, C., Wu, Q., Lee, Y.J.: Visual instruction tuning. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 34892–34916 (2023)
20. Nguyen, A.T., Nguyen, T.Q., Dang, D.L., Tran, D.T., Phan, B.D., Nguyen, Q.T., Le, T.: ViMoE-VQA: Reasoning-aware sparse experts for generative Vietnamese visual question answering. Procedia Computer Science (2026), to appear
21. Nguyen, A.T., Phan, B.D., Nguyen, T.Q., Tran, D.T., Dang, D.L., Nguyen, Q.T., Le, T.: AutoViVQA: A large-scale, automatically constructed dataset for Vietnamese visual question answering. In: Proceedings of the Asian Conference on Intelligent Information and Database Systems (ACIIDS). pp. 492–506. Communications in Computer and Information Science, Springer (2026)
22. Nguyen, N.S., Nguyen, V.S., Le, T.: Advancing Vietnamese visual question answering with transformer and convolutional integration. Computers and Electrical Engineering **119**, 109474 (2024)
23. Nguyen, Q.V., Tran, D.Q., Pham, H.Q., Nguyen, T.K.B., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViTextVQA: A large-scale visual question answering dataset for evaluating Vietnamese text comprehension in images (2025), arXiv preprint arXiv:2404.10652
24. Papineni, K., Roukos, S., Ward, T., Zhu, W.J.: BLEU: a method for automatic evaluation of machine translation. In: Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL). pp. 311–318 (2002)
25. Pham, H.Q., Nguyen, T.K.B., Nguyen, Q.V., Tran, D.Q., Nguyen, N.H., Nguyen, K.V., Nguyen, N.L.T.: ViOCRVQA: Novel benchmark dataset and vision reader for visual question answering by understanding Vietnamese text in images (2024), arXiv preprint arXiv:2404.18397
26. Qwen Team: Qwen2.5 technical report (2024), arXiv preprint arXiv:2412.15115
27. Satopää, V., Albrecht, J., Irwin, D., Raghavan, B.: Finding a "kneedle" in a haystack: Detecting knee points in system behavior. In: Proceedings of the International Conference on Distributed Computing Systems Workshops (ICDCSW). pp. 166–171 (2011)
28. Shang, Y., Cai, M., Xu, B., Lee, Y.J., Yan, Y.: LLaVA-PruMerge: Adaptive token reduction for efficient large multimodal models (2024), arXiv preprint arXiv:2403.15388

29. Tran, K.Q., Nguyen, A.T., Le, A.T.H., Nguyen, K.V.: ViVQA: Vietnamese visual question answering. In: Proceedings of the Pacific Asia Conference on Language, Information and Computation (PACLIC). pp. 683–691 (2021)
30. Tran, K.V., Nguyen, K.V., Nguyen, N.L.T.: BARTPhoBEiT: Pre-trained sequence-to-sequence and image transformers models for Vietnamese visual question answering. In: Proceedings of the International Conference on Multimedia Analysis and Pattern Recognition (MAPR). pp. 1–6 (2023)
31. Tran, K.V., Phan, H.P., Nguyen, K.V., Nguyen, N.L.T.: ViCLEVR: A visual reasoning dataset and hybrid multimodal fusion model for visual question answering in Vietnamese (2023), arXiv preprint arXiv:2310.18046
32. Tsimpoukelli, M., Menick, J., Cabi, S., Eslami, S.M.A., Vinyals, O., Hill, F.: Multimodal few-shot learning with frozen language models. In: Advances in Neural Information Processing Systems (NeurIPS). pp. 200–212 (2021)
33. Vedantam, R., Zitnick, C.L., Parikh, D.: CIDEr: Consensus-based image description evaluation. In: Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). pp. 4566–4575 (2015)
34. Yao, L., Li, L., Ren, S., Wang, L., Liu, Y., Sun, X., Hou, L.: DeCo: Decoupling token compression from semantic abstraction in multimodal large language models (2024), arXiv preprint arXiv:2405.20985